

Garvit Sharma

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EXPERIENCE

AHA Solar Technologies Limited

09/2023 – Present

Software Development Engineer

Ahmedabad, India

- Designed and scaled **distributed backend systems** serving **3.2M+ users**, ensuring **high availability**, horizontal scalability, and reliability at production scale.
- Architected **microservices** using **Node.js, Express.js, and MySQL** with **layered architecture (controller–manager–data layer)**, applying **separation of concerns** for modularity and testability.
- Built **RESTful APIs** with focus on **statelessness, idempotency, and fault tolerance**, supporting **15K+ daily requests** across **high-throughput, low-latency workflows**.
- Centralized **Joi-based** validation and middleware logic, reducing code duplication by **20%** across **15+ service endpoints**.
- Optimized performance via **Redis caching** and **query optimization**, cutting API response time by **60%** using **cache-aside patterns**.
- Automated **CI/CD** on **AWS EC2** with **S3** storage, enabling **zero-downtime deployments** and **99.9% uptime**.

PROJECTS

PMT-2.0 (AHASolar Project Management Tool)

02/2025 – Present

- Spearheaded **scalable, distributed backend architecture** for **20+ EPC companies**, achieving **99.9% uptime** and reducing project completion time by **15%**.
- Designed **200+ REST APIs** handling **2K+ daily requests** with **stateless, loosely coupled, modular service design** following **agile** practices.
- Enforced **JWT authentication** and **role-based access control (RBAC)** with fine-grained **authorization policies** across **200+ API endpoints**.
- Redesigned **MySQL schema** with **normalization and indexing strategies** across **20K+ records**, improving **query performance by 35%** and reducing **read latency**.
- Migrated to **Redis caching layer**, cutting **end-to-end latency by 60%** and improving **system throughput** under peak load.
- Automated **async event pipelines** using **cron** and **Docker**, improving **operational resilience** and reducing manual intervention by **30%**.

PMSG (Pradhan Mantri Surya Ghar) (Completed)

12/2023 – 02/2025

- Architected **fault-tolerant, horizontally scalable** backend for India's **10M-user national-scale platform**, applying **distributed systems principles** for reliability.
- Delivered **40+ APIs** across Vendor, DISCOM, and Consumer workflows (**10K+ daily requests**); built **React.js** dashboards with **role-based data isolation** for 3 user tiers.
- Implemented **JWT stateless authentication** with **tiered permission controls** and **access control enforcement** across **10M+ users**.
- Built **cron-based async data pipelines** with **batch processing** and **backpressure handling**, processing **450K+ records/month** and saving **30+ hours/month**.
- Established **data integrity and input validation pipelines** maintaining **95% data accuracy** and **data consistency** across **500K+ interactions**.

ACADEMIC PROJECTS

Indian Sign Language Recognition – OpenCV, TensorFlow, CNN

11/2022 – 06/2023

- Trained a **CNN-based deep learning model** on **25K+ annotated ISL images**, improving classification accuracy by **25%**; dataset on Kaggle with **60+ downloads**.
- Optimized **real-time inference pipeline** to **<200ms latency** via **efficient model serving**, enabling **500+ daily gesture recognitions**.
- Achieved **84.24%** accuracy across 50 participants; presented at **BITS Pilani Goa**; published with **Springer**.

EDUCATION

Vellore Institute of Technology

09/2022 – 05/2024

Master of Computer Applications (MCA) | **CGPA: 8.3**

Chennai, Tamil Nadu

SKILLS

Languages: JavaScript, SQL

Backend: Node.js, Express.js, FastAPI, RESTful API Design, JWT Authentication, RBAC, Cron, Input Validation, Microservices

Frontend: React.js, Redux

Cloud & DevOps: AWS (S3, EC2), Docker, CI/CD, Git, GitHub, GitLab

Databases: MySQL, PostgreSQL, Redis, Database Schema Design, Query Optimization, Indexing

Practices: Agile, Layered Architecture, Distributed Systems, System Design, High Availability, Scalability, Performance Optimization, Fault Tolerance, Cache-Aside Pattern, Async Processing